

YEAR 12
ASSESSMENT TASK 2
TERM 1, 2015

MATHEMATICS
EXTENSION 1

General Instructions:

- Reading Time: 5 minutes.
- Working Time: 2 hours.
- Write in black or blue pen.
- Board approved calculators & templates may be used
- A Standard Integral Sheet is provided.
- In Question 6 - 12, show all relevant mathematical reasoning and/or calculations.
- Marks may not be awarded for careless or badly arranged working.

Total Marks 82

Section I: 5 marks

- Attempt Question 1 – 5.
- Answer on the Multiple Choice answer sheet provided.
- Allow about 8 minutes for this section.

Section II: 77 Marks

- Attempt Question 6 - 12
- Answer on blank paper unless otherwise instructed. Start a new page for each new question.
- Allow about 1 hours & 52 minutes for this section.

Section 1 Multiple Choice (5 marks)

Attempt Questions 1 - 5 (1 mark each).

Allow approximately 8 minutes for this section

1 What is the domain and range of $y = \cos^{-1}\left(\frac{5x}{2}\right)$?

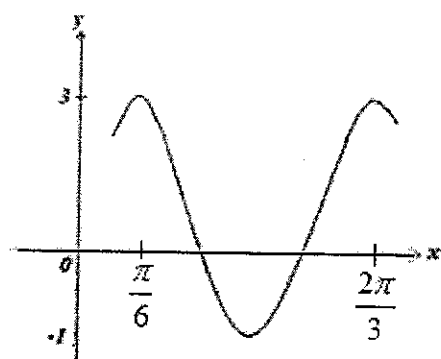
(A) Domain: $-\frac{2}{5} \leq x \leq \frac{2}{5}$, Range: $-\pi \leq y \leq \pi$

(B) Domain: $-1 \leq x \leq 1$, Range: $0 \leq y \leq \pi$

(C) Domain: $-1 \leq x \leq 1$, Range: $-\pi \leq y \leq \pi$

(D) Domain: $-\frac{2}{5} \leq x \leq \frac{2}{5}$, Range: $0 \leq y \leq \pi$

2 The graph below could have the equation



(A) $y = 2 \cos\left(x + \frac{\pi}{6}\right) + 1$

(B) $y = 2 \cos 4\left(x - \frac{\pi}{6}\right) + 1$

(C) $y = 2 \cos 2\left(x - \frac{\pi}{6}\right) + 1$

(D) $y = 2 \cos 4\left(x + \frac{2\pi}{3}\right) + 1$

3 What is the greatest coefficient in the expansion of $(5+2x)^{12}$?

(A) ${}^{12}C_3 \times 5^9 \times 2^3$

(B) ${}^{12}C_4 \times 5^8 \times 2^4$

(C) ${}^{12}C_5 \times 5^7 \times 2^5$

(D) ${}^{12}C_6 \times 5^6 \times 2^6$

4 What is the indefinite integral for $\int (\cos 3x - \sin^2 x + 2) dx$?

(A) $\frac{1}{3} \sin 3x - \frac{3}{2}x - \frac{1}{4} \sin 2x + c$

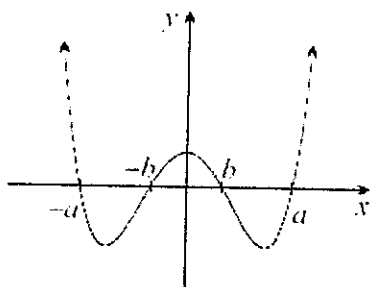
(B) $\frac{1}{3} \sin 3x - \frac{3}{2}x + \frac{1}{4} \sin 2x + c$

(C) $\frac{1}{3} \sin 3x + \frac{3}{2}x + \frac{1}{4} \sin 2x + c$

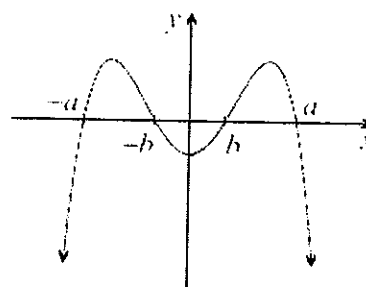
(D) $\frac{1}{3} \sin 3x + \frac{3}{2}x - \frac{1}{4} \sin 2x + c$

5. Which diagram best represents $P(x) = (x-a)^2(b^2-x^2)$, where $a > b$?

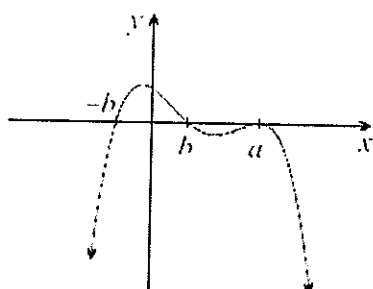
(A)



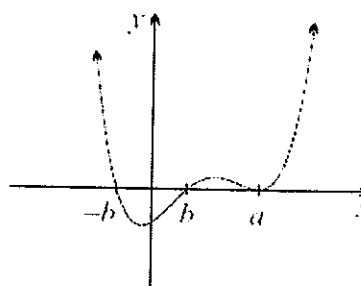
(B)



(C)



(D)



Section II Extended Response (77 marks)

Attempt Questions 6-12. Each question is worth 11 marks.

Allow approximately 1 hour 52 minute for this section.

Question 6 (Start a new page)

- (a) Solve $\log_2 x + \log_2(x^2 - 4) - \log_2(x + 2) = 3$. 3
- (b) Express $0.5\dot{1}\dot{2}$ as a geometric series and hence write its equivalent fraction in simplest term. 2
- (c) Use Simpson's approximation method with two sub-intervals to find an approximation to $\int_0^1 e^{x^2} dx$. Give your answer correct to 4 decimal places. 2
- (d) Sketch the graph of $y = 3 \cos^{-1} 2x$. 2
- (e) Evaluate $\cos\left[\sin^{-1}\left(\frac{-12}{13}\right)\right]$ exactly. 2

Question 7 (Start a new page)

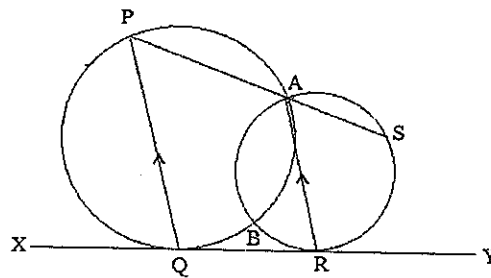
- (a) For the arbitrary integer p , find the sum of the series: 2
 $2^{2p-1} + 2^{2p-2} + 2^{2p-3} + \dots + 2^{-2p}$
- (b) Prove by induction that 5
 $3^n > 1 + 2n$ for $n > 1$
- (c) The curve $y = \sin 3x$, is rotated about the x axis. Find the exact volume of the solid of revolution from $x = 0$ and $x = \frac{\pi}{3}$. 4

Question 8 (Start a new page)

- (a) Find the following integral, $\int_0^{\frac{3\sqrt{3}}{2}} \frac{x dx}{\sqrt{9 - x^2}}$ using the substitution $x = 3 \sin \theta$. 4
- (b) Find the general solutions of the equation 3
 $\tan \theta = \sin 2\theta$
- (c) Find the stationary points on the curve $y = x^2 e^{-2x}$ and sketch the graph. 4

Question 9 (Start a new page)

- (a) Andy borrowed \$50000 to purchase a car at 6% per annum reducible interest, calculated monthly. The loan is to be repaid in 60 equal monthly instalments.
- (i) Calculate the monthly repayment he must pay to the nearest cent. 3
- (ii) With the twelfth repayment, Andy receives \$4000, from an aunt and pays this to the bank towards the repayments of his loan. 3
How many more repayments will be needed to pay off his loan? (assuming the monthly repayment has not changed)
- (b) Two circles meet at A and B. QR is a common tangent.



PQ is parallel to AR and PA is produced to meet the other circle at S.
Prove that:

- (i) QPSR is cyclic. 3
(ii) QA is parallel to RS. 2

Question 10 (Start a new page)

- (a) An hour glass has the shape of a double cone 30 cm high. The radii at both ends are 2 cm. Sand is flowing from the top cone into the bottom cone at the rate of $6 \text{ cm}^3/\text{s}$.

(i) Show that $r = \frac{2h}{15}$

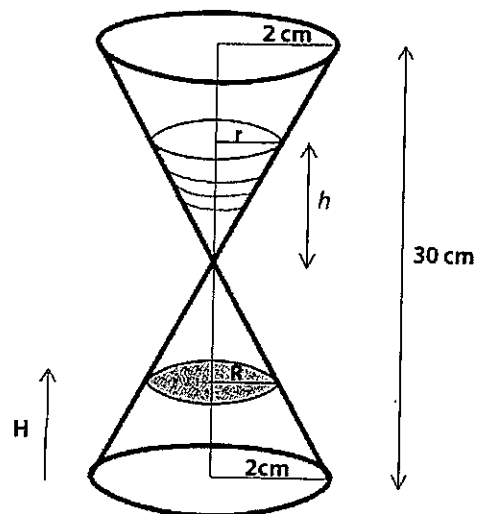
1

- (ii) At what rate is the level of the sand in the top cone falling when the coffee is 6 cm deep?

2

- (iii) How fast is the level of sand in the bottom cone rising at the instant when the sand in this cone is 6 cm deep?

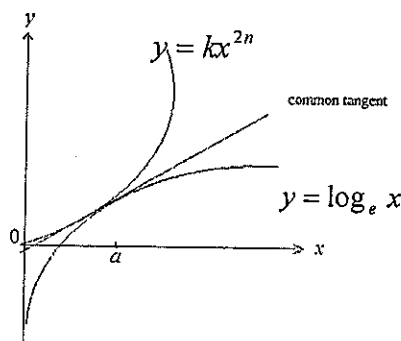
3



- (b) Two points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are on the parabola $x^2 = 4ay$.
- (i) The equation of the tangent to $x^2 = 4ay$ at an arbitrary point $(2at, at^2)$ on the parabola is $y = tx - at^2$. (Do not prove this.) Show that the tangents at points P and Q meet at R , where R is the point $(a(p+q), apq)$. 2
- (ii) As P varies, the point Q is always chosen so that $\angle POQ$ is a right angle, where O is the origin. Find the locus of R . 3

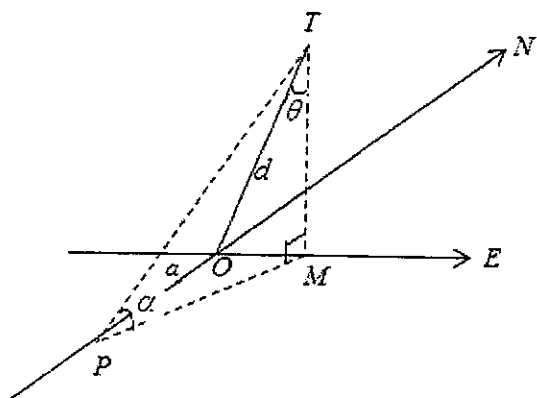
Question 11 (Start a new page)

(a)



The graphs of the functions $y = kx^{2n}$ and $y = \log_e x$ have a common tangent at $x = a$ as shown in the diagram.

- (i) By considering gradients, show that $a^{2n} = \frac{1}{2nk}$ 1
- (ii) Express k as a function of n by eliminating a . 3
- (b) A pole, OT , of length d metres stand on horizontal ground. The pole leans towards the east, making an angle θ with the vertical. From P , a metres south of O , the elevation of T is α .



(i) Prove that $PM = d \cos \theta \cot \alpha$. 2

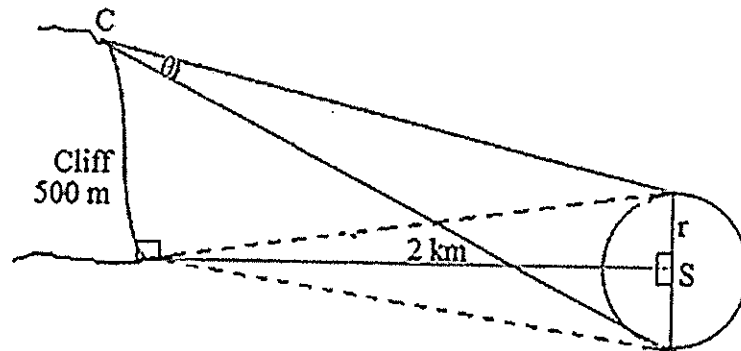
(ii) Prove that

$$d^2 = \frac{a^2}{\cos^2 \theta \cot^2 \alpha - \sin^2 \theta} \quad 2$$

- (c) Express $\cos x - \sqrt{3} \sin x$ in the form $R \cos(x + a)$, where $R > 0$ and $0 < x < \frac{\pi}{2}$ 3

Question 12 (Start a new page)

- (a) Find the value of x which is independent of x in the expansion $\left(2x^3 + \frac{1}{x^3}\right)^{12}$. 3
- (b) The sum of the cubes of three consecutive integers is given by $S_n = (n-1)^3 + n^3 + (n+1)^3$. Prove by method of induction that S_n is divisible by 3 for all $n \geq 1$. 4
- (c) A sky diver falls into the sea forming circular waves at S. An observer standing on the cliff (C) is measuring the circular waves formed at S from a position C, 500 metres above sea level and 2 kilometres horizontally from the centre of the waves. 4



- (i) At a certain time, the observer measures the angle θ , subtended by the diameter of the wave, to be 0.1 radians. What is the radius r , at this time? Give your answer correct to 1 decimal place. 2
- (ii) At this time $\frac{d\theta}{dt} = 0.01$ radians per hour. Find the rate at which the radius of the wave is growing. Give your answer correct to 1 decimal place. 2

END OF EXAMINATION

1. Domain $-\frac{1}{2} \leq x \leq 1$
 $= -\frac{2}{5} \leq x \leq \frac{2}{5}$
 Range is $0 \leq y \leq \pi$
 Answer (D)

2. Amplitude 2
 Cos graph moved one unit up
 and $\frac{\pi}{6}$ to right
 Answer (B)

3. $(5+2x)^{12}$
 $T_{r+1} = {}^{12}C_r 5^{12-r} (2x)^r$ and $T_r = {}^{12}C_{r-1} 5^{13-r} (2x)^{r-1}$
 $= {}^{12}C_r 5^{12-r} 2^r x^r$ $= {}^{12}C_{r-1} 5^{13-r} 2^{r-1} x^{r-1}$
 $\frac{T_{r+1}}{T_r} = \frac{{}^{12}C_r 5^{12-r} 2^r x^r}{{}^{12}C_{r-1} 5^{13-r} 2^{r-1} x^{r-1}}$
 ratio of terms

$$\frac{(13-r) \times 2}{r \times 5}$$

$$= \frac{26-2r}{5r}$$

$\therefore$ Greatest coefficient occurs when $r=3$
 $T_4 = {}^{12}C_3 \times 5^9 \times 2^3 \times x^3$
 has greatest coefficient when ${}^{12}C_3 \times 5^9 \times 2^3$
 Answer is (A)

4. $\int (\cos 3x - \sin^2 x + 2) dx$
 $= \int \left(\cos 3x - \frac{1}{2}(1 - \cos 2x) + 2 \right) dx$
 $= \frac{1}{3} \sin 3x - \frac{1}{2}x + \frac{1}{4} \sin 2x + 2x + C$
 $= \frac{1}{3} \sin 3x + \frac{3}{2}x + \frac{1}{4} \sin 2x + C$
 $\therefore$ Answer is (C)

5. It is a $-x^4$ graph
 parabola at $x=0$
 and x -intercepts at $x=b$ and $x=-b$.
 $\therefore$ graph is C

Question 6

(a) $\log_2 x + \log_2 (x^2 - 4) - \log_2 (x + 2) = 3$

$$\log_2 \frac{x(x^2 - 4)}{x + 2} = 3 \quad \checkmark$$

$$\log_2 \frac{x(x+2)(x-2)}{x+2} = 3$$

$$\log_2 x(x-2) = 3$$

$$2^3 = x^2 - 2x$$

$$8 = x^2 - 2x$$

$$0 = x^2 - 2x - 8 \quad \checkmark$$

$$0 = (x-4)(x+2)$$

$$x = 4 \text{ or } x = -2$$

$\therefore$ Only solution is $x = 4$ $\checkmark$

(b) $0.5121212 \dots$

$$0.5 + \frac{12}{1000} + \frac{12}{100000} \dots$$

$$0.5 + \underbrace{\frac{12}{1000} + \frac{12}{100000} + \dots}_{a = \frac{12}{1000}, r = \frac{1}{100}}$$

$$\therefore S_{\infty} = \frac{a}{1-r}$$

$$= \frac{12}{1000} \times \frac{100}{99}$$

$$= \frac{12}{990}$$

$$= \frac{4}{330} \quad \checkmark$$

$$\therefore \text{total is } \frac{1}{2} + \frac{4}{330} = \frac{169}{330} \quad \checkmark$$

(c)

x	0	$\frac{1}{2}$	1
e^{x^2}	$e^0 = 1$	$e^{\frac{1}{4}}$	e^1

Simpson's Rule

$$\frac{h}{3} [e^0 + 4 \times e^{\frac{1}{4}} + e^1] \quad \checkmark$$

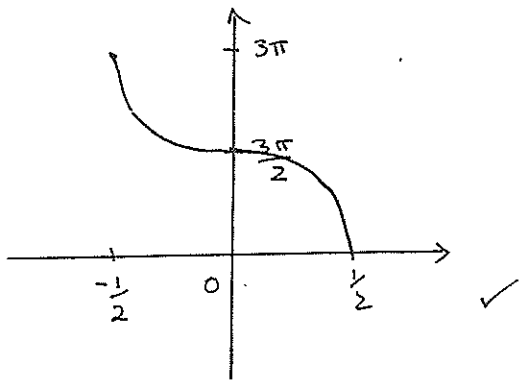
$$= \frac{0.5}{3} [1 + 5.136101667 + 2.718281828]$$

$$= \frac{0.5}{3} [8.854383495]$$

$$= 1.475730583$$

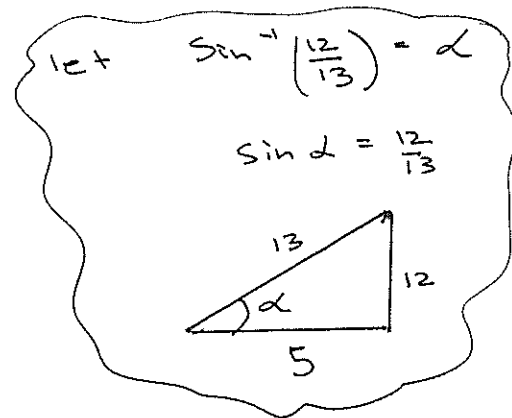
$$\approx 1.4757 \text{ sq units} \quad \checkmark$$

(9)



$$\left. \begin{array}{l} \text{Domain } -\frac{1}{2} \leq x \leq \frac{1}{2} \\ \text{Range } \pi < y \leq 3\pi \end{array} \right\} \checkmark$$

$$\begin{aligned} (e) \quad & \cos \left[\sin^{-1} \left(\frac{-12}{13} \right) \right] \\ &= \cos \left[-\sin^{-1} \left(\frac{12}{13} \right) \right] \\ &= \cos (-\alpha) \\ &= \cos \alpha \\ &= \frac{5}{13} \end{aligned}$$



Question 7

$$\begin{aligned} (a) \quad S_n &= 2^{2p-1} + 2^{2p-2} + \dots + 2^{-2p} \\ 2S_n &= 2^{2p} + 2^{2p-1} + 2^{2p-2} + \dots + 2^{-2p-1} \quad \checkmark \\ 2S_n - S_n &= 2^{2p} + 2^{-2p} \\ \therefore S_n &= 2^{2p} - 2^{-2p} \quad \checkmark \end{aligned}$$

OR

$$a = 2^{2p-1}, \quad r = 2^{-1} = \frac{1}{2} \quad \text{and} \quad l = 2^{-2p}$$

$$\begin{aligned} \therefore S_n &= \frac{a + rl}{1 - r} \\ &= \frac{2^{2p-1} - 2^{-1} \cdot 2^{-2p}}{1 - \frac{1}{2}} \quad \checkmark \\ &= \frac{2^{2p-1} - 2^{-2p-1}}{2^{-1}} \\ &= 2^{2p} - 2^{-2p} \quad \checkmark \end{aligned}$$

1(b) Prove $3^n > 1 + 2n$ for $n > 1$

Step 1 Prove $n=2$

$$3^2 - 2(2) - 1 > 0$$

$$9 - 4 - 1 > 0$$

true for $n=2$

✓

Step 2 Assume true for $n=k$

$$3^k - 2k - 1 > 0$$

Step 3 Prove true for $n=k+1$

✓

$$3^{k+1} - 2(k+1) - 1 > 0$$

LHS

$$= 3^{k+1} - 2k - 3$$

$$= 3 \cdot 3^k - 2k - 3$$

✓

$$3(3^k - 2k - 1) + 4k$$

which is > 0 since $3^k - 2k - 1 > 0$

✓

as is $4k$

Step 4 true for $n=k+1$ is true for $n=1$ and

true for all positive integral $n > 1$

$\therefore$ proved by mathematical induction

✓

7(c) $V = \pi \int_0^{\pi/3} \sin^2 3x \cdot dx$

✓

$$V = \pi \int_0^{\pi/3} \frac{1}{2} (1 - \cos 6x) \cdot dx$$

$$V = \frac{\pi}{2} \int_0^{\pi/3} (1 - \cos 6x) dx$$

✓

$$= \frac{\pi}{2} \left[x - \frac{1}{6} \sin 6x \right]_0^{\pi/3}$$

$$= \frac{\pi}{2} \left[\left(\frac{\pi}{3} - \frac{1}{6} \sin \frac{6\pi}{3} \right) - \left(0 - \frac{1}{6} \sin 0 \right) \right]$$

✓

$$= \frac{\pi}{2} \left[\left(\frac{\pi}{3} - \frac{1}{6} \sin 2\pi \right) - 0 \right]$$

$$= \frac{\pi}{2} \left[\frac{\pi}{3} \right]$$

$$= \frac{\pi^2}{6} \text{ unit}^3$$

✓

Question 8

$$\begin{aligned}
 \text{(a)} \quad \int_0^{\frac{3\sqrt{3}}{2}} \frac{x}{\sqrt{9-x^2}} dx &= \int_0^{\frac{\pi}{3}} \frac{3 \sin \theta \cdot 3 \cos \theta}{\sqrt{9-9\sin^2 \theta}} \\
 &= \int_0^{\frac{\pi}{3}} \frac{9 \sin \theta \cos \theta}{\sqrt{9 \cos^2 \theta}} \\
 &= \int_0^{\frac{\pi}{3}} \frac{9 \sin \theta \cos \theta}{3 \cos \theta} \\
 &= \int_0^{\frac{\pi}{3}} 3 \sin \theta \\
 &= \left[-3 \cos \theta \right]_0^{\frac{\pi}{3}} \\
 &= \left[-3 \times \frac{1}{2} + 3 \right] \\
 &= \frac{3}{2}.
 \end{aligned}$$

$$\begin{aligned}
 \text{let } x &= 3 \sin \theta \\
 dx &= 3 \cos \theta \\
 x &= \frac{\sqrt{3}}{2} \\
 \sin \theta &= \frac{\sqrt{3}}{2} \\
 \therefore \theta &= \frac{\pi}{3}
 \end{aligned}$$

$$\text{(b)} \quad \tan \theta = \sin 2\theta$$

$$\tan \theta - \sin 2\theta = 0$$

$$\frac{\sin \theta}{\cos \theta} - 2 \sin \theta \cos \theta = 0$$

$$\cancel{\cos \theta} \sin \theta - 2 \sin \theta \cos^2 \theta = 0$$

$$\sin \theta (1 - 2 \cos^2 \theta) = 0$$

$$\sin \theta = 0 \quad \text{or} \quad 1 - 2 \cos^2 \theta = 0$$

$$\sin \theta = \sin 0 \quad \cos^2 \theta = \frac{1}{2}$$

$$\therefore \theta = n\pi$$

$$\cos \theta = \pm \sqrt{\frac{1}{2}}$$

$$\cos \theta = \frac{1}{\sqrt{2}}$$

$$\text{or} \quad \cos \theta = -\frac{1}{\sqrt{2}}$$

$$\cos \theta = \cos \frac{\pi}{4}$$

$$\cos \theta = \cos \frac{3\pi}{4}$$

$$\therefore \theta = 2n\pi \pm \frac{\pi}{4}$$

$$\therefore \theta = 2n\pi \pm \frac{3\pi}{4}$$

8(c) $y = x^2 e^{-2x}$

stationary points occur when $y' = 0$

Product rule

let $u = x^2 \therefore u' = 2x$

$v = e^{-2x} \therefore v' = -2e^{-2x}$

$\therefore y' = vu' + uv'$

$= e^{-2x} \cdot 2x + x^2 \cdot -2e^{-2x}$

$= 2xe^{-2x} - 2x^2 e^{-2x}$

$= 2xe^{-2x} (1 - x)$

when $2xe^{-2x} = 0$ or $1 - x = 0$

$x = 0$

$x = 1$

when $x = 0$, $y = 0 \therefore$ stationary point at $(0, 0)$

when $x = 1$, $y = e^{-2} \therefore$ stationary point at $(1, e^{-2})$

Nature either second derivative or test points on either side of first derivative:

x	$-\frac{1}{2}$	0	$\frac{1}{2}$
y''	$-\frac{3e}{2}$	///	$\frac{1}{2e}$

-ve

+ve

✓ Minimum point

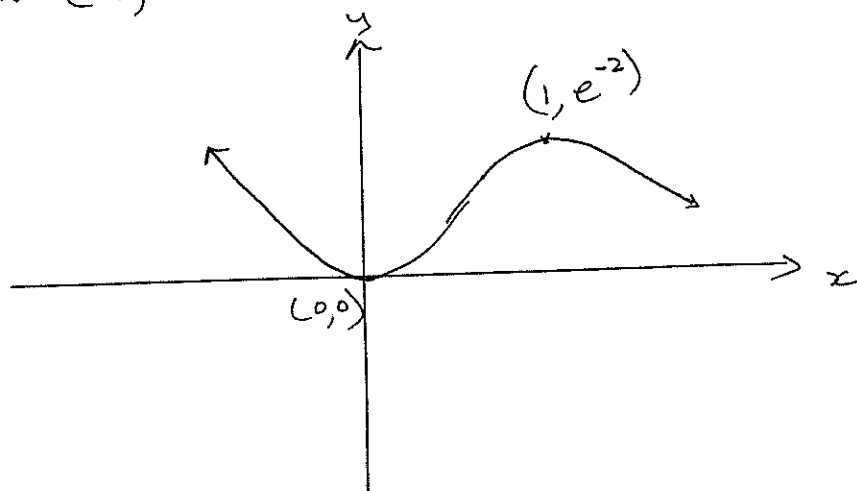
x	$\frac{1}{2}$	1	$\frac{3}{2}$
y''	$\frac{1}{2e}$	///	$-\frac{3e^3}{2}$

+ve

-ve

∧ Maximum point

$\therefore$ at $(0, 0)$ - there is minimum point
and at $(1, e^{-2})$ there is a maximum point.



Question 9
(a) (i) \$50000

Interest = 6% = 0.5% per month
= 0.005

paid in 60 monthly installments

we know
 $A_1 = 50000 \times 1.005^{-n}$
 $A_n = 50000 \times 1.005^{-n} - M(1 + 1.005 + 1.005^2 + \dots + 1.005^{n-1})$
 $A_n = 0$
 $\therefore 0 = 50000 \times 1.005^{-n} - M(1 + 1.005 + 1.005^2 + \dots + 1.005^{n-1})$

$$S(n) = \frac{a(r^n - 1)}{r - 1}$$

$$= \frac{1(1.005^n - 1)}{1.005 - 1}$$

$$= \frac{1.005^n - 1}{0.005}$$

$$= 69.77003051$$

$$\therefore M = \frac{50000 \times 1.005^n}{69.77003051}$$

$$= 966.6460765$$

$$= \$966.64$$

(ii) $A_{12} = 50000 \times 1.005^{-12} - 966.64(1.005^{12} - 1)$

$$= 38325.95199$$

$$= \$38325.95$$

An extra \$4000 paid means $(38325.95 - 4000) = \$34325.95$ remains to be paid off.

$$\therefore 34325.95 \times 1.005^n - 966.64(1.005^n - 1) = 0$$

$$34325.95 \times 1.005^n - 966.64 + 966.64(1.005^n) = 0$$

$$34325.95 \times 1.005^n + 966.64(1.005^n) - 966.64 = 0$$

$$34325.95 \times 1.005^n + 966.64(1.005^n) = 966.64$$

$$34325.95 \times 1.005^n = 966.64 - 966.64(1.005^n)$$

$$\therefore 1.005^n = \frac{966.64}{34325.95 + 966.64}$$

$$= \frac{966.64}{35292.59}$$

$$= 0.02739$$

$$1.005^n = \frac{966.64}{35292.59}$$

$$\therefore 1.005^n = 0.02739$$

$$\log 1.005 = \log 0.02739$$

$$n \log 1.005 = \log 0.02739$$

$$\therefore n = \frac{\log 0.02739}{\log 1.005}$$

$$= \frac{-1.5606}{0.002231437}$$

$$= -700.0$$

Therefore there are 70 more payments left.

9(b)(i) $\angle MPQ = \angle KAR$ (corresponding angles as $PQ \parallel AR$) ✓

$\angle RAS = \angle SRY$ (angle between the chord and tangent is equal to any angle in the alternate segment) ✓

$\therefore \angle APQ = \angle SRY$

$\therefore QPSR$ is cyclic (the interior angle in a cyclic quadrilateral is equal to the opposite exterior angle.) ✓

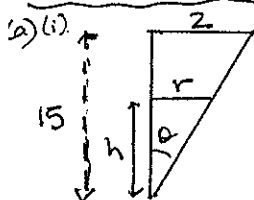
(ii) $\angle AQR = \angle APQ$ (interior angle in a cyclic quadrilateral is equal to the opposite exterior angle) ✓

from part (i)

$\angle AQR = \angle SRY$

$QA \parallel RS$ (corresponding angles in parallel lines) ✓

Question 10



(i) $\begin{cases} \tan \theta = \frac{2}{15} \\ \tan \theta = \frac{r}{h} \end{cases}$

$\frac{r}{h} = \frac{2}{15}$

$\therefore r = \frac{2h}{15}$ ✓

or (1) $\frac{r}{2} = \frac{h}{15}$ + (corresponding sides of similar triangles are proportional.)
 $\therefore r = \frac{2h}{15}$

no explanation = 0 marks

(ii) $\frac{dV}{dt} = -6 \text{ cm}^3/\text{s}$ and $h = 6 \text{ cm}$.

$$\begin{aligned} V_{\text{cone}} &= \frac{1}{3} \pi r^2 h \\ &= \frac{1}{3} \pi \left(\frac{2h}{15} \right)^2 h \\ &= \frac{4\pi h^3}{3 \times 225} \end{aligned}$$

$$\Rightarrow \frac{dV}{dh} = \frac{12\pi h^2}{3 \times 225} = \frac{4\pi h^2}{225} \quad (1)$$

$$\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$$

$$-6 = \frac{4\pi (6^2)}{225} \times \frac{dh}{dt}$$

$$\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$$

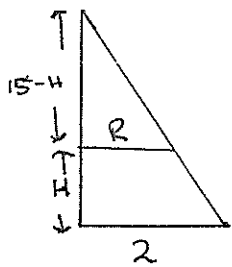
$$\frac{dh}{dt} = \frac{225}{4\pi (6^2)} \times -6$$

$$\therefore \frac{dh}{dt} = \frac{-6 \times 225}{4\pi (36)}$$

$$= -\frac{75}{8\pi}$$

$\therefore$ level is falling at the rate of $\frac{75}{8\pi} \text{ cm/s}$ (1)

For bottom cone - similarity



$$\frac{R}{2} = \frac{15-H}{15}$$

$$\therefore R = \frac{2}{15}(15-H)$$

Amount of sand is:

$$V = \frac{1}{3} \pi r^2 h - \frac{1}{3} \pi R^2 h$$

$$= \frac{\pi}{3} (4) \cdot 15 - \frac{\pi}{3} R^2 (15-H)$$

$$= 20\pi - \frac{\pi}{3} \left(\left(\frac{2}{15}(15-H) \right)^2 (15-H) \right)$$

$$= 20\pi - \frac{\pi}{3} \times \frac{4}{225} (15-H)^3$$

$$= 20\pi - \frac{4\pi}{3 \times 225} (15-H)^3$$

$$\frac{dV}{dH} = \frac{3 \times 4\pi (15-H)^2}{3 \times 225}$$

$$= \frac{4\pi (15-H)^2}{225}$$

$$\frac{dV}{dt} = 6$$

$$\boxed{\frac{dV}{dt} = \frac{dV}{dH} \times \frac{dH}{dt}}$$

$$6 = \frac{4\pi (15-H)^2}{225} \times \frac{dH}{dt}$$

$$6 = \frac{4\pi (15-6)^2}{225} \times \frac{dH}{dt}$$

$$6 = \frac{4\pi (9^2)}{225} \times \frac{dH}{dt}$$

$$6 = \frac{324\pi}{225} \cdot \frac{dH}{dt}$$

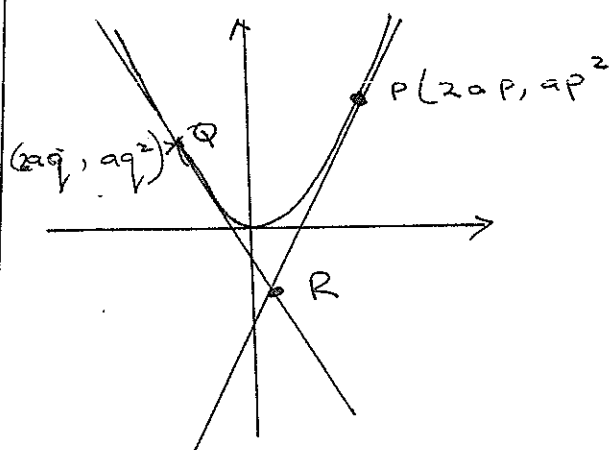
$$\therefore \frac{dH}{dt} = \frac{6 \times 225}{324\pi}$$

$$= \frac{25}{6\pi}$$

$\therefore$ Sand is rising at the rate of

$$\frac{25}{6\pi} \text{ cm/s} \approx 1.326 \text{ cm/s}$$

Q. 10 (b)(i)



From equation $y = t(x - at^2)$.

tangent of P is

$$y = px - ap^2$$

tangent of Q is

$$y = qx - aq^2$$

R is the intersection

$$px - ap^2 = qx - aq^2$$

$$px - qx = ap^2 - aq^2$$

$$x(p - q) = a(p^2 - q^2)$$

$$x(p - q) = a(p + q)(p - q)$$

$$\therefore x = a(p + q)$$

(1)

when $x = a(p + q)$ then $y = p(a(p + q)) - ap^2$

$$ap(p + q) - ap^2$$

$$= ap^2 + apq - ap^2$$

$$= apq$$

(1)

$\therefore R$ is $(a(p + q), apq)$

b(ii) If $\angle POQ = 90^\circ$

then $M_{PO} = \frac{ap^2 - 0}{2ap - 0} = \frac{p}{2}$

and $M_{QO} = \frac{aq^2 - 0}{2aq - 0} = \frac{q}{2}$

(1)

then $M_{PO} \times M_{QO} = -1$

$$\frac{p}{2} \times \frac{q}{2} = -1$$

$$pq = -4$$

(1)

at R $x = a(p + q)$

$$y = apq$$

$$y = -4a$$

$\therefore$ locus of R is $y = -4a$.

(1)

a(i) $y = kx^{2n}$ then $y' = 2nkx^{2n-1}$
 $y = \log_e x$ then $y' = \frac{1}{x}$

at point a both have same gradient

$$2nkx^{2n-1} = \frac{1}{x}$$

at point a

$$\left. \begin{aligned} 2nka^{2n-1} &= a^{-1} \\ a^{2n-1} &= \frac{1}{2nka} \\ a^{2n} \cdot a^{-1} &= \frac{1}{2nka} \\ \therefore a^{2n} &= \frac{1}{2nk} \end{aligned} \right\}$$

(ii) when graphs intersect at a

$$ka^{2n} = \log_e a$$

we know $a^{2n} = \frac{1}{2nk}$

add logs on both sides ✓

$$2n \log_e a = -\log_e 2nk$$

$$2n \cdot ka^{2n} = -\log_e 2nk$$

$$2nk \cdot \frac{1}{2nk} = -\log_e 2nk$$

$$1 = -\log_e 2nk$$

$$e^{-1} = 2nk$$

$$k = \frac{1}{2ne}$$

we know $\log_e a = ka^{2n}$

Alternatively $ka^{2n} = \log_e a$

$$k \cdot \frac{1}{2nk} = \log_e a$$

$$\frac{1}{2n} = \log_e a$$

$$e^{-2n} = a$$

Substitute a

$$(e^{-2n})^{2n} = \frac{1}{2nk}$$

$$e^{-4n^2} = \frac{1}{2nk}$$

$$k = \frac{1}{2n} \times e^{4n^2}$$

$$k = \frac{e^{4n^2}}{2n}$$

(b) (i) $\sin \theta = \frac{OM}{d} \quad \therefore OM = d \sin \theta$
 $\cos \theta = \frac{TM}{d} \quad \therefore TM = d \cos \theta$ } ✓
 $\tan \alpha = \frac{TM}{PM}$
 $\therefore PM = \frac{TM}{\tan \alpha}$
 $= TM \cot \alpha$ ✓
 $= d \cos \theta \cot \alpha$

(ii) $a^2 = PM^2 - OM^2$
 $a^2 = (d \cos \theta \cot \alpha)^2 - (d \sin \theta)^2$ ✓
 $a^2 = d^2 \cos^2 \theta \cot^2 \alpha - d^2 \sin^2 \theta$
 $a^2 = d^2 (\cos^2 \theta \cot^2 \alpha - \sin^2 \theta)$
 $\therefore d^2 = \frac{a^2}{\cos^2 \theta \cot^2 \alpha - \sin^2 \theta}$ ✓

11 c $\cos x - \sqrt{3} \sin x = R \cos(x + a)$
 $= R \cos x \cos a - R \sin x \sin a$ ✓
 $R \cos a = 1$ and $R \sin a = +\sqrt{3}$ and $R^2 = 4$
 $\tan a = \sqrt{3} \quad \therefore R = 2$
 $\therefore a = \frac{\pi}{3}$ ✓
 $\therefore \cos x - \sqrt{3} \sin x = 2 \cos\left(x + \frac{\pi}{3}\right)$ ✓

Question 12

a). $T_{r+1} = {}^{12}C_r a^{12-r} b^r$ for $(a+b)^{12}$
For $(2x^3 + x^{-3})^{12}$
 $T_{r+1} = {}^{12}C_r (2x^3)^{12-r} (x^{-3})^r = {}^{12}C_r (2^{12-r} x^{36-3r} x^{-3r})$
 $= {}^{12}C_r \cdot 2^{12-r} \cdot x^{36-6r}$ ✓
 k is independent of x (ie k is a constant term)
 $= k x^0$
 $\therefore x^{36-6r} = x^0$ ✓

$36-6r=0$
 $r=6$
 $T_n = {}^{12}C_7 2^{12-7} = 2^5 {}^{12}C_7 = 32 \cdot \frac{12!}{5!7!} = 25344$ ✓

Divisible by 3

$$= (n-1)^3 + n^3 + (n+1)^3 = 3k \quad \text{where } k \text{ and } n \text{ are integers}$$

Step 1 for $n=1$

$$\text{LHS} = 0^3 + 1^3 + 2^3 = 9$$

$$\text{RHS} = 3k = 3 \times 3 = 9 \quad \text{so true for } n=1$$

Step 2 Assume S_n is true for $n=k$.

$$S(k) = (k-1)^3 + k^3 + (k+1)^3 = 3k$$

$$k^3 + (k+1)^3 = 3k - (k-1)^3$$

Step 3 prove for $n=k+1$

$$k^3 + (k+1)^3 + (k+2)^3 = 3k - (k-1)^3 + (k+2)^3$$

$$= 3k - (k^3 - 3k^2 + 3k - 1) + k^3 + 6k^2 + 12k + 8$$

$$= 3k - k^3 + 3k^2 - 3k + 1 + k^3 + 6k^2 + 12k + 8$$

$$= 3k + 9k^2 + 9k + 9$$

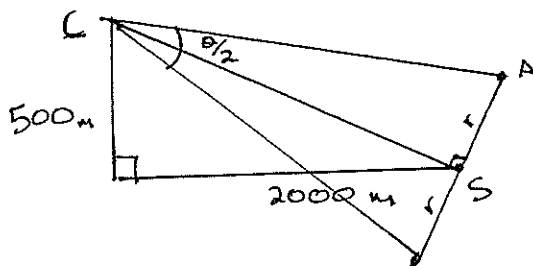
$$= 3(k + 3k^2 + 3k + 3)$$

$$= 3P \quad \text{where } P = k + 3k^2 + 3k + 3$$

$\therefore S(n)$ is true for $k+1$

Step 4 Since S_n is true for $n=1$, it is true for $n=2, n=3$ and so on
By MI S_n is true for all $n=1, 2, \dots$

12 c (i).



$$CS^2 = 500^2 + 2000^2$$

$$CS = 2052.08$$

$$= 2061.552813$$

In ΔCAS

$$\tan\left(\frac{\theta}{2}\right) = \frac{r}{2061.552813}$$

$$\therefore r = \tan\left(\frac{\theta}{2}\right) \times 2061.552813$$

$$= \tan\left(\frac{0.1}{2}\right) \times 2061.552813$$

$$= \tan 0.05 \times 2061.552813$$

$$= 103.6776407$$

$$= 103.1$$

$$\therefore r = 103.1 \text{ m}$$

If $\theta = 0.1$

$$\frac{d\theta}{dt} = 0.01$$

$$\boxed{\frac{dr}{dt} = \frac{dr}{d\theta} \cdot \frac{d\theta}{dt}}$$

$$= 1031 \text{ Sec}^2 (0.05) \times 0.01$$

$$= 10.31000078$$

$$10.3 (1 \text{ dp})$$

we know $r = \tan \frac{\theta}{2} \times 2062$

$$\frac{dr}{d\theta} = \left(2062 \sec^2 \frac{\theta}{2} \times \frac{1}{2} \right)$$

$$= 1031 \sec^2 \frac{\theta}{2}$$

we know $\frac{d\theta}{dt} = 0.02 \times 0.01$

$\therefore$ radius of waves is growing at a rate of 10.3 m/hr